Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience designing and maintaining infrastructure for data and ML pipelines, CI/CD, and cloud platforms using Python, Java, Apache Spark, AWS, and Kubernetes. Strong in big-data processing and SQL.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C/C++, SQL, Go.
Big Data & Pipelines: Apache Spark, PySpark, Delta Lake, Hadoop-style batch processing, medallion (bronze/silver/gold) pipelines, ETL.
Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs.
Databases: PostgreSQL, MySQL, DynamoDB, Redis.
Developer Tools: Git, Jenkins, GitLab CI/CD, AWS (S3, EC2, Lambda, SQS), Docker, Kubernetes, ArgoCD, Helm, Terraform, Ansible, Linux, Bash, SonarQube.
AI & Data: scikit-learn, PyTorch, TensorFlow, pandas, NumPy, Gemini, Vertex AI, LangChain.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Built config-driven data pipeline decoupling video-to-memory extraction from device automation for Android GUI bug reproduction, evaluated against the **ViBR** baseline across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) and designed a versioned run-artifact schema keeping every trial reproducible.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Designed and maintained **GitOps CI/CD** infrastructure with GitLab CI/CD and ArgoCD, enabling automated, auditable, rollback-capable deployments across 40+ production environments on AWS.
- Automated **Teamcenter X** infrastructure workflows in Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Analyzed live infrastructure performance across Linux, EC2, and databases with **CloudWatch** and **Datadog**, executing **44 production upgrades** in 4 months.
- Migrated **40+ environments** to Kubernetes with ArgoCD, Helm, Amazon EFS, and HashiCorp Vault; served as Quality Coach raising automated code coverage from **35% to 78%** via CI/CD quality gates.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built ML pipeline forecasting NYC taxi demand across **263 zones** with **PySpark**, Apache Spark, and Delta Lake, processing trip and weather data through a medallion pipeline into a **2.3M-row** hourly dataset.
- Designed a two-stage Poisson hurdle model cutting RMSE **43%** over baseline Random Forest; served predictions via a **FastAPI** REST service backed by PostgreSQL and Redis.

PolicyPilot – Policy Intelligence Portal [React, TS, Node, Gemini, MCP, Docker] May 2026

- Built a PDF-to-structured-data pipeline (parse, clean, chunk, validate) with schema validation at every boundary and Vitest tests; added citation and version tracing so stale-data mismatches surface instead of silently misleading a user.

EXTRACURRICULAR ACTIVITIES

Finalist, RevolutionUC 2025 Hackathon (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).